1. Why are loops necessary in programming?

Loops are necessary because they allow us to repeat a block of code multiple times without writing the same code again and again.

For example, if we want to print numbers from 1 to 5, instead of writing five print() statements, we can use a loop.

```
for i in range(1, 6):
    print(i)
```

Loops make the program shorter, easier to understand, and more efficient.

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2. What is the purpose of the pass keyword?

The pass keyword is used when we want to leave a block of code empty for now. It does nothing when the program runs.

It is useful when we are creating a function, loop, or condition but want to write the code later.

Example:

```
for i in range(5):
    pass
```

Here, the loop runs, but pass does not perform any action.

Another example:

```
def my_function():
    pass
```

This allows us to create the function without adding any code to it yet.

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3. What is the difference between break and continue?

Both break and continue are used inside loops, but they work differently.

break completely stops the loop.

Example:

```
for i in range(1, 6):
    if i == 3:
        break
    print(i)
```

Output:

1

2

When i becomes 3, the loop stops completely.

continue skips the current iteration and moves to the next iteration.

Example:

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

Output:

1

2

4

5

Here, only 3 is skipped, while the loop continues.

difference:

break → stops the entire loop
continue → skips the current iteration

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4. What is the difference between while loop and for loop?

Both are used to repeat code, but they are generally used in different situations.

for loop	while loop
Used when we know what we want to iterate over or have a predictable sequence/count.	Used when repetition depends on a condition.
Commonly used with sequences and range().	Continues as long as the condition is True.
Example: for i in range(5)	Example: while age < 18

Example of for loop:

```
for i in range(5):
    print(i)
```

Example of while loop:

```
i = 0


while i < 5:
    print(i)
    i += 1
```

Both examples print:

0

1

2

3

4

In simple words:

For loop → when we have a sequence or know the number/range of iterations.
While loop → when we want to continue repeating until a condition becomes false.

4 points to remember:
Loop = repeat, **pass = do nothing**, **break = stop**, **continue = skip**.